Math Kangaroo Practice

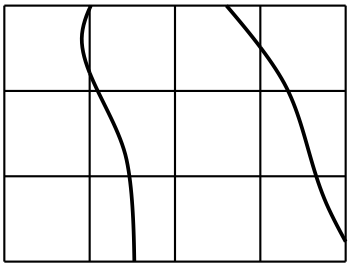
Grade 3-4 · 24 questions · standard · seed 20261021

Part A: Each question is worth 3 points

1. Five ribbons have these lengths.
Which ribbon is the longest?

A. 1 m 60 cm ***B.*** 15 dm ***C.*** 180 cm ***D.*** 2100 mm ***E.*** 25 dm

2. Leo painted a river and some roads on a map made of squares. Here is the whole picture.



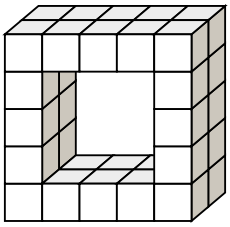
Which of these squares is **not** a piece of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

3. The odometer of a car shows 75249 km. All the digits of this number are different.
After how many more kilometres will all the digits be different again?

A. 21 ***B.*** 11 ***C.*** 10 ***D.*** 9 ***E.*** 1

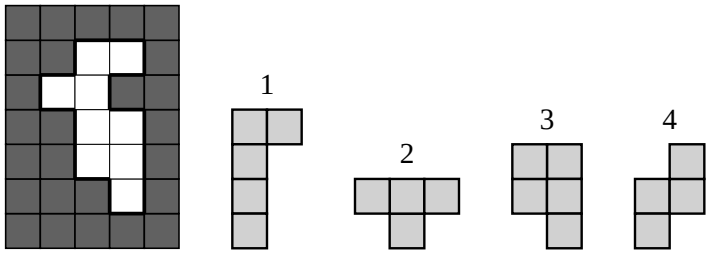
4. Robert built this tunnel out of identical cubes. The hole goes straight through from the front to the back.



How many cubes did Robert use?

A. 16 ***B.*** 24 ***C.*** 25 ***D.*** 32 ***E.*** 48

5. Which two of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.

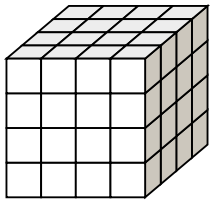


A. 1+2 ***B.*** 1+3 ***C.*** 1+4 ***D.*** 2+3 ***E.*** 3+4

6. Mia, who is older than Sara by 1 year and 1 day, was born on January 1, 2002. When was Sara born?

A. January 2, 2003 ***B.*** December 31, 2002 ***C.*** December 31, 2000
D. January 3, 2003 ***E.*** January 1, 2003

7. A big cube made of $4 \times 4 \times 4 = 64$ small cubes is painted on the outside, also underneath, and then taken apart.



How many small cubes have exactly 3 painted faces?

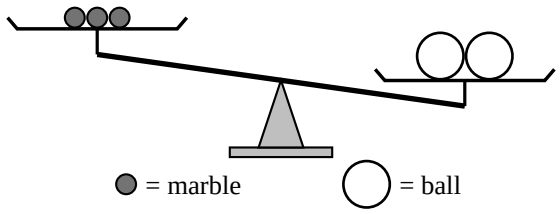
A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

8. A bar code is made of 17 black and white bars, black and white taking turns. The first and the last bar are black. Each black bar is either wide or narrow. There are 7 more white bars than wide black bars.

How many narrow black bars are there?

A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 2 ***E.*** 1

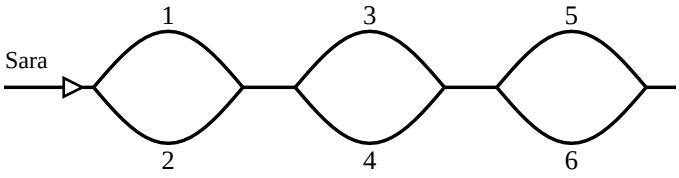
9. On one plate of a balance there are 3 marbles and on the other there are 2 balls. When we put one more ball, exactly like the others, on the plate with the marbles, the balance is level.



One ball weighs the same as how many marbles?

A. 1 marble ***B.*** 2 marbles ***C.*** 3 marbles ***D.*** 4 marbles ***E.*** 6 marbles

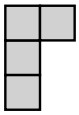
10. Sara walks along the road from left to right and writes down every number on the way.

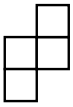
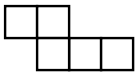
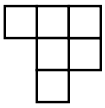
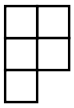


How many different groups of numbers can Sara write?

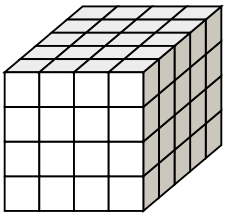
A. 8 ***B.*** 9 ***C.*** 10 ***D.*** 11 ***E.*** 12

11. Which of the five pieces fits together with the grey piece to make a rectangle? Pieces may be turned and flipped over.



A.  ***B.***  ***C.*** ***D.***  ***E.*** 

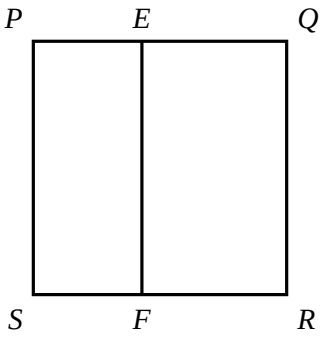
12. Tom built this brick (4 cubes long, 5 cubes deep and 4 cubes tall) from red and blue cubes of the same size. Every cube you can see from the outside (also from below and behind) is red. All the cubes inside are blue.



How many red cubes did Tom use?

A. 68 ***B.*** 74 ***C.*** 80 ***D.*** 84 ***E.*** 112

13. *PQRS* is a square. Its side is 7 cm. *PEFS* is a rectangle. Its shorter side is 3 cm.



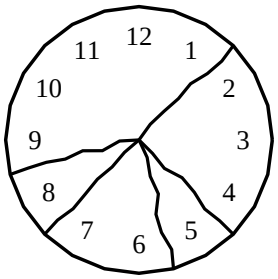
How many centimetres is the perimeter of the square *PQRS* larger than that of the rectangle *PEFS*?

A. 16 cm ***B.*** 8 cm ***C.*** 6 cm ***D.*** 5 cm ***E.*** 4 cm

14. To write all the seat numbers in a theatre, starting from 1, exactly 29 digits are needed. How many seats are there?

A. 29 ***B.*** 28 ***C.*** 22 ***D.*** 19 ***E.*** 11

15. The face of a clock is cracked into 5 pieces as shown.



Which piece has the smallest sum of numbers?

A. the piece with the 4 ***B.*** the piece with the 8 ***C.*** the piece with the 12
D. the piece with the 5 ***E.*** the piece with the 7

Part B: Each question is worth 4 points

16. Which word has exactly 1 different letter in common with GIRAFFE?

A. PLANET ***B.*** ELEPHANT ***C.*** BANANA ***D.*** SANDWICH ***E.*** PROBLEM

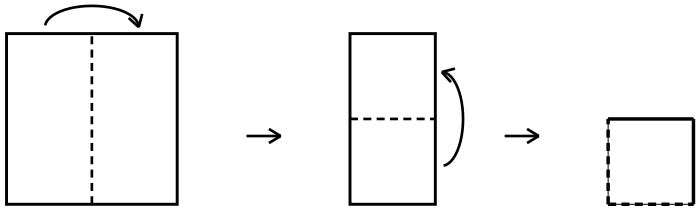
17. What is the biggest number smaller than 512 whose sum of digits is the same as for 512?

A. 546 ***B.*** 521 ***C.*** 511 ***D.*** 503 ***E.*** 502

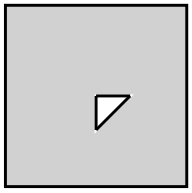
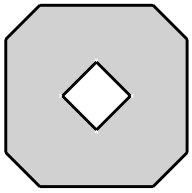
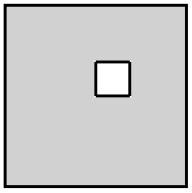
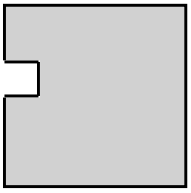
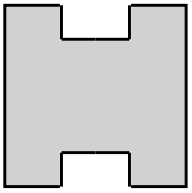
18. All the numbers from 1 to 100 are written down.
How many times is the digit 0 written?

A. 9 ***B.*** 10 ***C.*** 11 ***D.*** 12 ***E.*** 13

19. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). One corner of the small square is cut off, and the paper is unfolded.



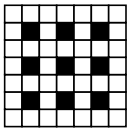
Which design **can** be made this way?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

20. The diagrams below are the 1st, 2nd and 3rd of a pattern. How many white cells will the 5th diagram have?

8 white cells

21 white cells


40 white cells

A. 133 ***B.*** 121 ***C.*** 96 ***D.*** 75 ***E.*** 65

21. Sofia, Fabienne, Aisha and Chloe each play a different instrument: the guitar, the flute, the violin and the drum. Chloe plays an instrument with strings. Aisha blows into the instrument. Fabienne uses a bow.
Which statement is true?

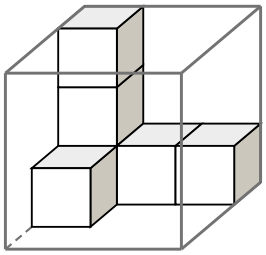
A. Chloe plays the guitar ***B.*** Fabienne plays the guitar ***C.*** Chloe plays the flute
D. Aisha plays the guitar ***E.*** Fabienne plays the drum

22. Martha has three cards with different digits. Two of them are 6 and 1; the third is face down. Using each card exactly once, with +, –, ×, ÷ and brackets, Martha gets 21.
Which of these digits can be on the hidden card?

A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 4 ***E.*** 3

Part C: Each question is worth 5 points

23. Sara puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Sara wants the two bottom layers to be completely full.



How many more cubes are needed for that?

A. 13 ***B.*** 14 ***C.*** 15 ***D.*** 16 ***E.*** 21

24. Anna paints a flag with three stripes: top, middle and bottom. Anna has 3 colours (white, green and yellow). All three stripes must be different colours.
How many different flags can Anna paint?

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 27

Answer Key

1.	**E**	**2.**	**B**	**3.**	**B**	**4.**	**D**	**5.**	**E**	**6.**	**A**
7.	**A**	**8.**	**B**	**9.**	**C**	**10.**	**A**	**11.**	**E**	**12.**	**A**
13.	**B**	**14.**	**D**	**15.**	**D**	**16.**	**C**	**17.**	**D**	**18.**	**C**
19.	**E**	**20.**	**C**	**21.**	**A**	**22.**	**E**	**23.**	**A**	**24.**	**A**

Solutions

1. E (25 dm) — In cm: 1 m 60 cm = 160 cm, 15 dm = 150 cm, 180 cm = 180 cm, 2100 mm = 210 cm, 25 dm = 250 cm. The longest is 25 dm. [2003 Q10 · longest]

2. B — Find each square in the picture by the places where its lines reach the edges. Squares A, C, D, E can be found; square B is a piece mirrored and upside down, so it is not in the picture like that. [2002 Q1 · not_a_piece]

3. B (11) — Counting up, every number before 75260 repeats a digit. 75260 has all different digits: 75260 – 75249 = 11. [2002 Q24 · all_different]

4. D (32) — The front has 5 × 5 – 3 × 3 = 16 cubes. The tunnel is 2 cubes long, so 16 × 2 = 32. [2002 Q20 · count_tunnel]

5. E (3+4) — The empty area has 9 squares. Pieces have 1: 5, 2: 4, 3: 5, 4: 4 squares. Only pieces 3 and 4 also fit together in the right shape. [2003 Q18 · two_pieces]

6. A (January 2, 2003) — Older means born earlier, so Sara was born 1 year and 1 day **after** January 1, 2002: 1 year later is January 1, 2003, then 1 more day: January 2, 2003. [2007 Q12 · younger_birthday]

7. A (8) — Only the corner cubes have 3 painted faces, and a cube always has 8 corners, however big it is. [2007 Q17 · always_eight]

8. B (8) — Black is at both ends, so there are 9 black and 8 white bars. Wide: 8 – 7 = 1. Narrow: 9 – 1 = 8. [2003 Q17 · narrow_bars]

9. C (3 marbles) — Take 1 ball off both plates: 3 marbles balance 2 – 1 = 1 ball. So one ball = 3 marbles. [2002 Q9 · one_more_balances]

10. A (8) — The forks have 2, 2, 2 branches; every choice can go with every other: 2 × 2 × 2 = 8. [2007 Q1 · how_many_routes]

11. E — The grey piece and the right piece together fill a 3 × 3 rectangle: 9 – 4 = 5 squares, in exactly the shape of the gap. Piece E (turned) fills it; the others have a square too many, too few, or in the wrong place. [2007 Q8 · complete_rectangle]

12. A (68) — All cubes: 4 × 5 × 4 = 80. Blue inside: 2 × 3 × 2 = 12. Red: 80 – 12 = 68. [2003 Q9 · red_outside]

13. B (8 cm) — Square: 4 × 7 = 28 cm. Rectangle: 3 + 7 + 3 + 7 = 20 cm. Difference 28 – 20 = 8 cm. [2002 Q6 · square_minus_strip]

14. D (19) — 1 to 9 use 9 digits. The other 29 – 9 = 20 digits belong to two-digit numbers: 20 ÷ 2 = 10 numbers, from 10 to 19. So there are 19. [2003 Q16 · pages_from_digits]

15. D (the piece with the 5) — The sums are: 2+3+4 = 9, 5 = 5, 6+7 = 13, 8 = 8, 9+10+11+12+1 = 43. [2002 Q11 · biggest_piece]

16. C (BANANA) — BANANA shares A with GIRAFFE. SANDWICH shares 2; PROBLEM shares 2; ELEPHANT shares 2; PLANET shares 2. [2007 Q3 · which_word]

17. D (503) — 5 + 1 + 2 = 8. Counting down from 512, the first number with digit sum 8 is 503. [2007 Q4 · before_same_sum]

18. C (11) — A 0 at the end: 10, 20, ..., 100: 10 numbers. Plus the zeros in the middle of 100-100 (like 101 or 100 itself): 1 more. Total 11. [2007 Q23 · count_zeros]

19. E — A single cut appears four times, the same on the left and right and on the top and bottom. Only design E looks like that (notches in the middle of the top and bottom sides). [2007 Q24 · which_can]

20. C (96) — Diagram number k is (2k + 1) × (2k + 1) with k × k black cells. The 5th: 11 × 11 – 5 × 5 = 121 – 25 = 96. [2007 Q21 · far_diagram]

21. A (Chloe plays the guitar) — The only possible matching: Sofia – drum; Fabienne – violin; Aisha – flute; Chloe – guitar. [2002 Q18 · which_true]

22. E (3) — (1 + 6) × 3 = 21. The other digits do not work with 6 and 1. [2002 Q21 · hidden_card]

23. A (13) — Two layers hold 2 × 3 × 3 = 18 cubes. In those two layers there are already 5 cubes (columns count at most 2 each). 18 – 5 = 13. [2007 Q14 · two_layers]

24. A (6) — Top stripe: 3 choices, middle: 2, bottom: 1. 3 × 2 × 1 = 6. [2002 Q22 · stripes]